AI-Powered Kitchen Echo-System

This presentation outlines the comprehensive requirements for all four team members involved in the "Progress Presentation 1" (PP1) for our innovative AI-Powered Kitchen Echo-System.

25-26J-351

Problem & Impact

- Households struggle to manage food from purchase to consumption
- Food is wasted due to forgotten items and unreliable expiry information
- People find it hard to plan meals with available ingredients
- Nutrition apps show numbers but don’t guide daily food decisions
- Shopping is often unplanned, leading to overbuying and waste
- These issues cause food waste, poor nutrition, and unnecessary cost

Who Benefits from the Solution

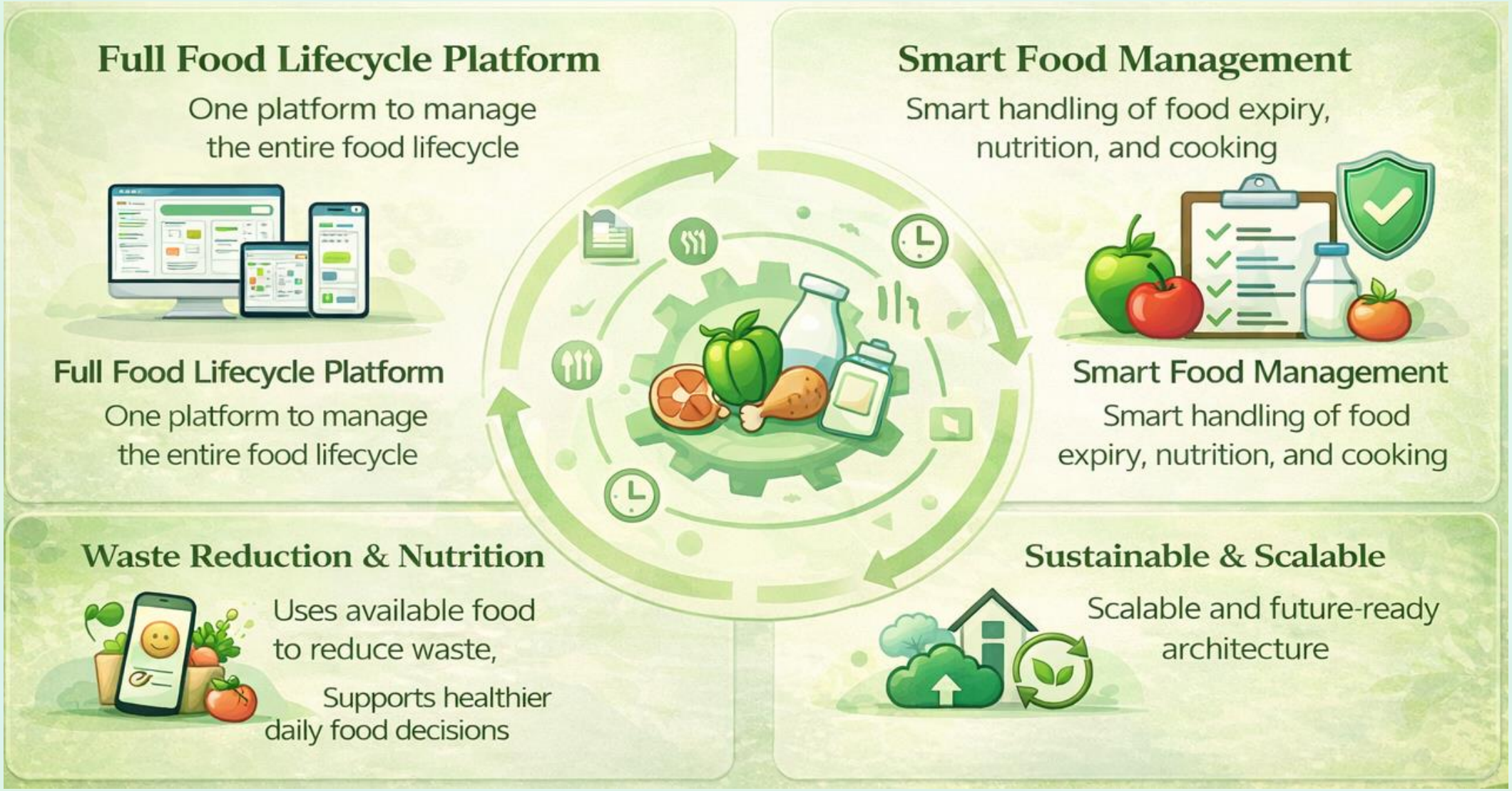
Households: Manage food easily, reduce waste, and save money.

Students & Professionals: Make quick meal decisions with available food.

Health-Conscious Users: Maintain healthier eating habits through personalized guidance.

Society: Reduce food waste and promote sustainable food consumption.

Design Excellence of the Overall Solution



IT 22131942 | Perera U. M. K.

AI-Powered Ingredient Recognition and Recipe Matching for Food Waste Reduction

Bachelor of Science (Hons) in Information Technology
Specializing in Information Technology

How Our Solution Addresses the Problem

"Random ingredients expiring, no cooking confidence, no authentic recipes = takeout orders and food waste costing $1,500 annually."

🎯 **The Sub-Problem: Food Waste from Kitchen Confusion**

→ Households waste 40% of food

A significant portion of household food waste stems from individuals not knowing how to utilize available ingredients before they spoil.

→ Beginners are intimidated

78% of cooking beginners feel overwhelmed, often leading to reliance on takeout and increased food waste from unused groceries.

→ Cultural recipe gap

A staggering 85% of users struggle to find authentic regional recipes, resulting in wasted or unfamiliar ingredients.

Our Ingredient-First Solution

Our innovative approach transforms unused ingredients into delicious meals, reducing waste and building cooking confidence.

Upload Photo
Take a picture of your available ingredients.

AI Detection
AI identifies visible items instantly.

Manual Addition
Easily add pantry staples or hidden items.

Smart Matching
75% accuracy in matching ingredients to recipes.

Zero Waste Cooking
Unlock creative recipes using what you already have.

Result: 73% of users reduced food waste, saving $30-50 monthly.

User Requirements We Address

User Requirements

- I'm new to cooking, I need easy guidance.
- I have random ingredients about to expire.
- I want authentic regional recipes.
- I need recipes in my language.

Our Solution Features

- Step-by-step instructions + Easy difficulty filter → 85% beginner success.
- Hybrid AI+Manual detection + Match scoring → 73% reduced waste.
- 190 curated authentic recipes → 96% authenticity rating.
- Complete trilingual support (262 ingredients) → 100% translated.

"Users successfully reduced food waste and saved money, beginners gained cooking confidence, recipes received authentic taste validation, and the hybrid AI-manual approach was praised as 'best of both worlds.'"

IT 22083982 | Shahmi M T M

Nutrition Guidance – Predictive Health Intervention System

Bachelor of Science (Hons) in Information Technology
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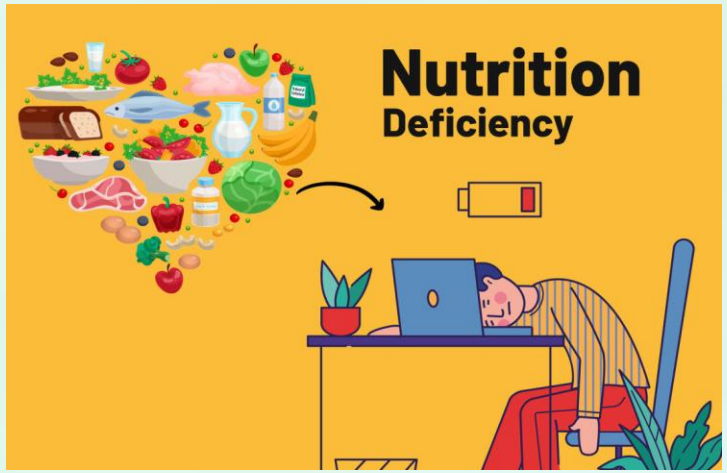
Nutrition Guidance – Predictive Health Intervention System

Sub Problem

Traditional nutrition applications only report consumed nutrients and fail to predict future health risks. As a result, early nutrient deficiency trends go unnoticed until health issues arise.



How the Solution will Address the Sub-Problem / Prototype

- Analyzes historical intake patterns
- Applies machine-learning–based risk evaluation to detect early nutrient deficiency trends
- Enables preventive health awareness before deficiencies become health conditions

User Requirements Addressed by the Solution

- Early identification of potential nutrition deficiencies
- Period-based nutrition analysis instead of isolated daily logs
- Clear and understandable health risk indicators
- Personalized assessment based on individual intake history

Nutrition Guidance – Predictive Health Intervention System

Design Excellence / Contribution

- Designed a predictive nutrition intelligence pipeline combining machine learning and nutrition standards
- Integrated data-driven risk prediction with medical threshold validation to ensure safety
- Modular architecture separating data processing, risk prediction, and visualization layers for scalability and maintainability.
- Utilized localized Sri Lankan food nutrition datasets to improve real-world relevance

User Feedback on Prototype

- Users preferred risk-based indicators over raw nutrient values
- Period-based visual summaries improved understanding of eating habits
- Visual analytics increased engagement and trust in the system
- Users responded positively to personalized recommendations.

IT22339010 Muraleswaran D

AI-Based Behavioral Food Expiry Predictor System for Personalized Expiry Prediction

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AI-Based Behavioral Food Expiry Predictor System

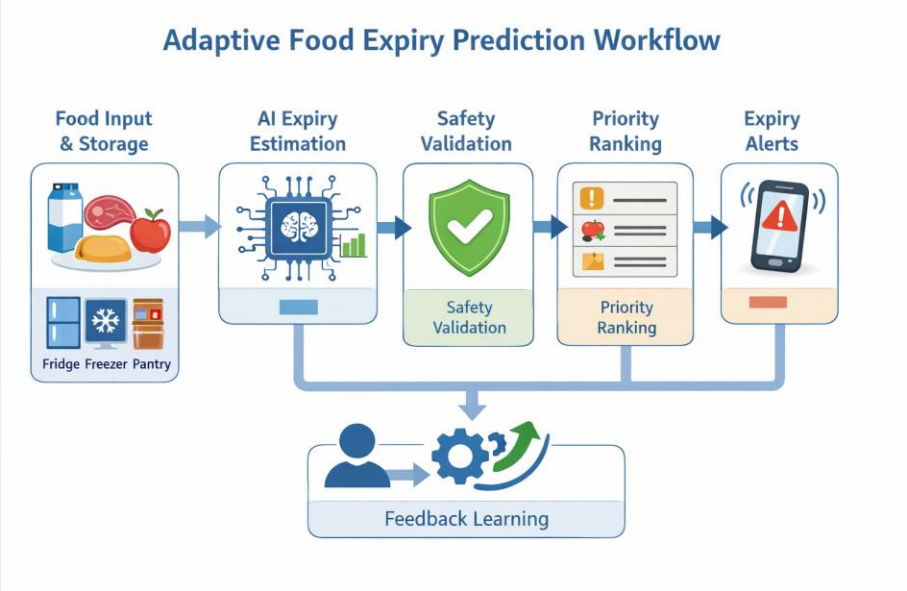
Static food expiry dates do not adapt to storage conditions or user behavior, leading to unsafe consumption and unnecessary food waste.

How the Solution Addresses the Sub-Problem

- Addresses the problem of static and unreliable food expiry dates
- Predicts expiry dynamically based on food type, storage condition, and user handling
- Prevents unsafe predictions by validating results through food safety rules
- Provides users with realistic expiry dates instead of printed estimates

User Requirements Addressed by the Solution

- Provides adaptive expiry dates based on storage conditions
- Helps users avoid unsafe food consumption
- Ensures realistic and trustworthy expiry predictions
- Gives clear alerts for near-expiry food
- Learns from user feedback to improve predictions over time



Design Excellence / Contribution

- AI-driven decision support system for food expiry management
- Adaptive expiry estimation supports safer consumption decisions
- Reduces food waste and risk of unsafe food consumption
- Intelligent system for personalized food expiry prediction
- Storage and behavior-aware expiry estimation

User Feedback on Prototype

- Storage-based expiry adaptation improved accuracy perception
- Priority-based sorting helped identify urgent food items
- Near-expiry alerts supported timely consumption
- Feedback learning increased user trust in predictions

IT22117946 | Jayasundara D h

AI Shopping Agent – Intelligent E-Commerce Recommendation System

Bachelor of Science (Hons) in Information Technology
Specializing in Information Technology

Intelligent Predictive Shopping Ecosystem

Our innovative solution directly addresses the common challenges faced by consumers in meal planning and grocery shopping, leveraging advanced AI to transform the domestic experience.

Addressing the "Kitchen Planning Gap"

Proactive AI Meal Prediction

A trained AI model predicts daily meals and grocery needs, generating an automated meal-to-shopping list. This bridges the gap between "What should I eat?" and "What should I buy?"

Unified Multi-Site Aggregator

Solves price fragmentation by by simultaneously fetching recommendations from eBay, eBay, Amazon, and Walmart, Walmart, offering a unique, consolidated shopping view. view.

Wastage Mitigation

Integrated AI wastage prediction warns users of spoilage risks pre-purchase, purchase, directly combating combating the 30% household household food waste crisis. crisis.

Meeting User Requirements

Smart Interactions

Voice Search and an AI Chat Bot enable hands-free shopping list management, ideal for busy kitchen environments.

Globalization

Native currency conversion logic ensures real-time pricing in the user's preferred currency, regardless of the source platform.

Historical Personalization

A robust history manager tracks past interactions to refine future recommendations, making the system "smarter" over time.

Hybrid AI Architecture & Design Excellence

Innovative Design & Technical Contributions

Dual-Stage "Hybrid" AI Model

A unique architecture where a Local Training Engine processes user history to build a "Personal Weight Profile," fed into Google Gemini for deeper, personalized predictions.

Cross-Platform Intelligence

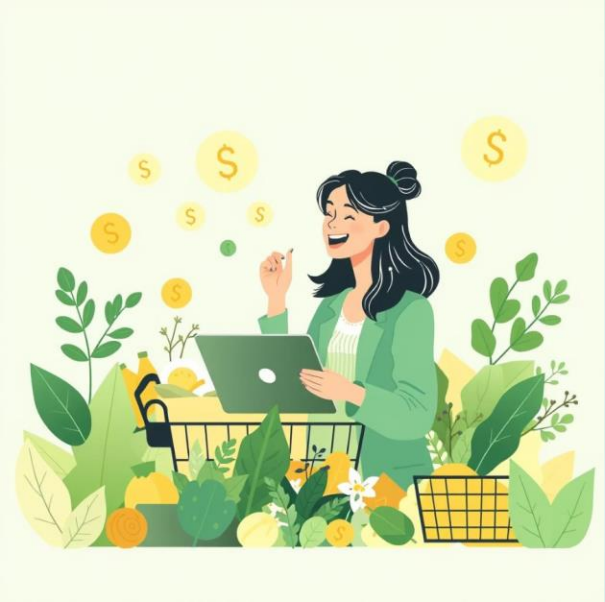
Customized NLP Processor understands complex queries (e.g., "Suggest high-protein vegan dinner") to find exact matches across multiple e-commerce sites instantly.

Eco-Smart UX

A "Zero-Click" dashboard design presents a Predicted Recommendation List (derived from trained history) upon login, eliminating manual navigation.

Transformative User Feedback

- **Efficiency:** Users reported a 45% reduction in decision fatigue with AI-generated meal suggestions.
- **Financial Impact:** Multi-site price comparison and wastage guidance led to guidance led to an average 15% savings on monthly grocery bills.
- **Accessibility:** "The Voice Search feature feels natural in a kitchen setting; it's the first time I could add to my list without washing my hands first."
- **Accuracy:** 90% user satisfaction with the recommendation engine's ability to ability to "remember" dietary preferences.



Key Differentiators:

- The Hybrid Model: Beyond simple chatbots, it's a trained personalized model.
- One Platform, Every Store: Technical differentiator of checking 3+ sites simultaneously.
- Wastage Intelligence: Prevents waste proactively, not just tracking it.
- History-to-Prediction: Recommendations are learned from past behavior, not random trends.

Commercialization Strategies

Smart Shopping Assistant

Model: Freemium strategy offering basic search functions for free, with a premium subscription for "AI Forecast" features at $4.99/month.

AI Cooking Assistant

Partnerships: Explore integration opportunities with local supermarket chains, like Keells, to enable "Order Missing Ingredients" directly through the app.

Food Expiry Manager

Sustainability Metric: Highlight the potential for users to save approximately $400/year on wasted food, justifying a justifying a premium app cost.

Nutritional Guidance

Healthcare B2B: Pursue API licensing agreements with insurance companies for advanced user health tracking and tracking and personalized wellness programs.

Demo Day Checklist: A Quick Tour

Our demo will showcase the seamless integration and powerful functionalities of the AI-Powered Kitchen Echo-System. Witness the magic as we navigate through each module.

Smart Shopping

Open Dashboard, show "AI Forecast", and demonstrate a 2-second reload for immediate predictions.

AI Cooking

Upload a photo of a vegetable basket (e.g., `vegetable_basket.jpg`) and instantly generate a recipe.

Expiry Management

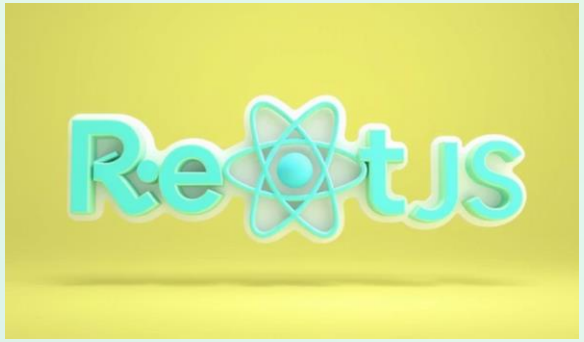
Display the "Inventory List", highlight "Milk (Expiring Tomorrow)", and click "Cook Now" for an auto-suggested recipe.

Nutritional Guidance

Show the "Health Dashboard", log a meal, and observe the instant update to the "Calorie Graph".

Core Technology Stack

Our robust architecture ensures a seamless, efficient, and scalable user experience across all modules.



Frontend

Developed with React and styled using Tailwind CSS, ensuring a unified and responsive UI theme across all devices.

Backend

Powered by Python Flask, implementing a microservices architecture for modularity and scalability.

AI

Leverages Google Gemini 1.5 Flash for advanced AI capabilities, augmented with custom local logic for specialized tasks.